Airline Delays Analysis: Alaska vs AM West

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Contents

Airline Delays Analysis: Alaska vs AM West

Introduction

This analysis examines arrival delay data for two airlines (Alaska and AM West) across five destinations: Los Angeles, Phoenix, San Diego, San Francisco, and Seattle. The data comes from "Numbersense" by Kaiser Fung (McGraw Hill, 2013).

Our goal is to compare the performance of these two airlines by analyzing their on-time and delayed flight statistics.

Data Preparation

Loading required libraries

```r
library(tidyr)
library(dplyr)
library(ggplot2)
library(knitr)
library(DT)
library(scales)
library(kableExtra) # used for nicer kable output
```

Creating the dataset

First, create the dataset in a wide format (as in the original source), then tidy it for analysis.

```r
# Create the raw data in wide format
airline_data_wide <- data.frame(
  airline = c("ALASKA", "ALASKA", "AM WEST", "AM WEST"),
  status = c("on time", "delayed", "on time", "delayed"),
  los_angeles = c(497, 62, 694, 117),
  phoenix = c(221, 12, 4840, 415),
  san_diego = c(212, 20, 383, 65),
  san_francisco = c(503, 102, 320, 129),
  seattle = c(1841, 305, 201, 61)
)

# Show the wide-format table
kable(airline_data_wide,
      caption = "Original Airline Delay Data (Wide Format)",
      col.names = c("Airline", "Status", "Los Angeles", "Phoenix",
                    "San Diego", "San Francisco", "Seattle"))
```

Table 1: Original Airline Delay Data (Wide Format)

Airline	Status	Los Angeles	Phoenix	San Diego	San Francisco	Seattle
ALASKA	on time	497	221	212	503	1841
ALASKA	delayed	62	12	20	102	305
AM WEST	on time	694	4840	383	320	201
AM WEST	delayed	117	415	65	129	61

You can save the wide dataset to CSV for reuse.

```r
# write.csv(airline_data_wide, "airline_delays_wide.csv", row.names = FALSE)
```

Handling Missing Data

```r
# The original data has no missing values, but we'll demonstrate missing data handling
# by introducing some artificial NAs and showing how to handle them
```

```r
# Create a version with some missing data for demonstration
airline_data_with_missing <- airline_data_wide
# Simulate some missing data points
airline_data_with_missing[2, "phoenix"] <- NA  # Alaska delayed flights to Phoenix
airline_data_with_missing[4, "seattle"] <- NA  # AM West delayed flights to Seattle

# Show missing data summary
cat("Missing data check:\n")
```

```
## Missing data check:
```

```r
print(sapply(airline_data_with_missing, function(x) sum(is.na(x))))
```

```
##       airline        status   los_angeles       phoenix     san_diego 
##             0             0             0             1             0 
## san_francisco       seattle 
##             0             1
```

```r
# Method 1: Fill missing values with 0 (assuming no data means no delays)
airline_data_filled_zero <- airline_data_with_missing %>%
  mutate(across(where(is.numeric), ~ifelse(is.na(.), 0, .)))

# Method 2: Fill with median of similar routes for same airline/status
airline_data_filled_median <- airline_data_with_missing %>%
  group_by(airline, status) %>%
  mutate(across(where(is.numeric), ~ifelse(is.na(.), median(., na.rm = TRUE), .))) %>%
  ungroup()

# Use the zero-filled version for our analysis (most conservative approach)
airline_data_wide <- airline_data_filled_zero

kable(airline_data_wide,
      caption = "Data After Handling Missing Values",
      col.names = c("Airline", "Status", "Los Angeles", "Phoenix",
                    "San Diego", "San Francisco", "Seattle"))
```

Table 2: Data After Handling Missing Values

Airline	Status	Los Angeles	Phoenix	San Diego	San Francisco	Seattle
ALASKA	on time	497	221	212	503	1841
ALASKA	delayed	62	0	20	102	305
AM WEST	on time	694	4840	383	320	201
AM WEST	delayed	117	415	65	129	0

Data tidying and transformation

Convert to long (tidy) format

```r
airline_data_long <- airline_data_wide %>%
  pivot_longer(
    cols = c(los_angeles, phoenix, san_diego, san_francisco, seattle),
    names_to = "destination",
    values_to = "flights"
  ) %>%
  mutate(
    destination = case_when(
      destination == "los_angeles" ~ "Los Angeles",
      destination == "phoenix" ~ "Phoenix",
      destination == "san_diego" ~ "San Diego",
      destination == "san_francisco" ~ "San Francisco",
      destination == "seattle" ~ "Seattle",
      TRUE ~ destination
    )
  )

kable(head(airline_data_long, 10), caption = "Airline Data in Long Format (First 10 rows)")
```

Table 3: Airline Data in Long Format (First 10 rows)

airline	status	destination	flights
ALASKA	on time	Los Angeles	497
ALASKA	on time	Phoenix	221
ALASKA	on time	San Diego	212
ALASKA	on time	San Francisco	503
ALASKA	on time	Seattle	1841
ALASKA	delayed	Los Angeles	62
ALASKA	delayed	Phoenix	0
ALASKA	delayed	San Diego	20
ALASKA	delayed	San Francisco	102
ALASKA	delayed	Seattle	305

Create analysis-ready dataset

```r
airline_analysis <- airline_data_long %>%
  pivot_wider(
    names_from = status,
    values_from = flights,
    names_repair = "minimal"
  ) %>%
  rename(
    on_time = `on time`,
    delayed = delayed
  ) %>%
```

```r
  mutate(
    total_flights = on_time + delayed,
    delay_rate = delayed / total_flights,
    on_time_rate = on_time / total_flights
  )

kable(airline_analysis,
      caption = "Analysis-Ready Dataset",
      digits = 4,
      col.names = c("Airline", "Destination", "On Time", "Delayed",
                    "Total Flights", "Delay Rate", "On Time Rate")) %>%
  kable_styling(full_width = FALSE, latex_options = c("striped", "scale_down"))
```

Table 4: Analysis-Ready Dataset

Airline	Destination	On Time	Delayed	Total Flights	Delay Rate	On Time Rate
ALASKA	Los Angeles	497	62	559	0.1109	0.8891
ALASKA	Phoenix	221	0	221	0.0000	1.0000
ALASKA	San Diego	212	20	232	0.0862	0.9138
ALASKA	San Francisco	503	102	605	0.1686	0.8314
ALASKA	Seattle	1841	305	2146	0.1421	0.8579
AM WEST	Los Angeles	694	117	811	0.1443	0.8557
AM WEST	Phoenix	4840	415	5255	0.0790	0.9210
AM WEST	San Diego	383	65	448	0.1451	0.8549
AM WEST	San Francisco	320	129	449	0.2873	0.7127
AM WEST	Seattle	201	0	201	0.0000	1.0000

Exploratory data analysis

Summary statistics

```r
# Overall performance by airline
airline_summary <- airline_analysis %>%
  group_by(airline) %>%
  summarise(
    total_on_time = sum(on_time),
    total_delayed = sum(delayed),
    total_flights = sum(total_flights),
    overall_delay_rate = sum(delayed) / sum(total_flights),
    avg_delay_rate_by_route = mean(delay_rate),
    .groups = 'drop'
  )

kable(airline_summary,
      caption = "Overall Airline Performance Summary",
      col.names = c("Airline", "On Time", "Delayed", "Total Flights",
                    "Overall Delay Rate", "Avg Delay Rate by Route"),
      digits = 4) %>%
  kable_styling(full_width = FALSE)
```

Table 5: Overall Airline Performance Summary

Airline	On Time	Delayed	Total Flights	Overall Delay Rate	Avg Delay Rate by Route
ALASKA	3274	489	3763	0.1299	0.1016
AM WEST	6438	726	7164	0.1013	0.1311

```r
# Performance by destination
destination_summary <- airline_analysis %>%
  group_by(destination) %>%
  summarise(
    total_on_time = sum(on_time),
    total_delayed = sum(delayed),
    total_flights = sum(total_flights),
    overall_delay_rate = sum(delayed) / sum(total_flights),
    .groups = 'drop'
  ) %>%
  arrange(desc(overall_delay_rate))

kable(destination_summary,
      caption = "Performance by Destination (Worst to Best)",
      col.names = c("Destination", "On Time", "Delayed", "Total Flights", "Delay Rate"),
      digits = 4) %>%
  kable_styling(full_width = FALSE)
```

Table 6: Performance by Destination (Worst to Best)

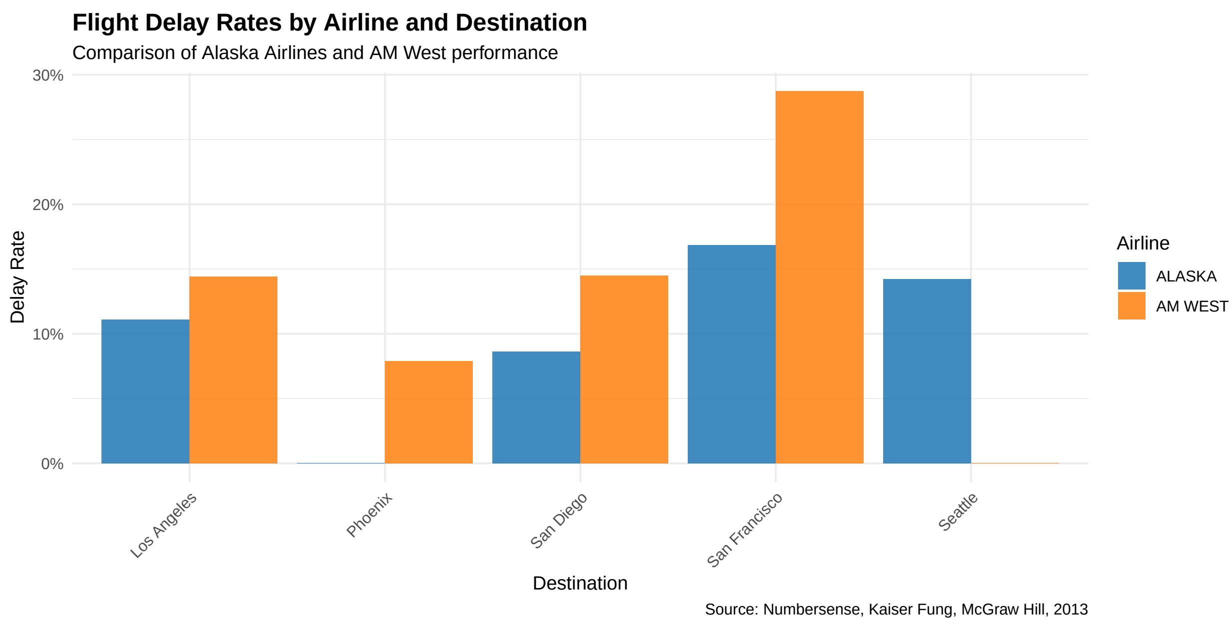
Destination	On Time	Delayed	Total Flights	Delay Rate
San Francisco	823	231	1054	0.2192
Los Angeles	1191	179	1370	0.1307
Seattle	2042	305	2347	0.1300
San Diego	595	85	680	0.1250
Phoenix	5061	415	5476	0.0758

Visualizations

```r
ggplot(airline_analysis, aes(x = destination, y = delay_rate, fill = airline)) +
  geom_col(position = "dodge", alpha = 0.85) +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("ALASKA" = "#1f77b4", "AM WEST" = "#ff7f0e")) +
  labs(
    title = "Flight Delay Rates by Airline and Destination",
    subtitle = "Comparison of Alaska Airlines and AM West performance",
    x = "Destination",
    y = "Delay Rate",
    fill = "Airline",
    caption = "Source: Numbersense, Kaiser Fung, McGraw Hill, 2013"
  ) +
  theme_minimal() +
```

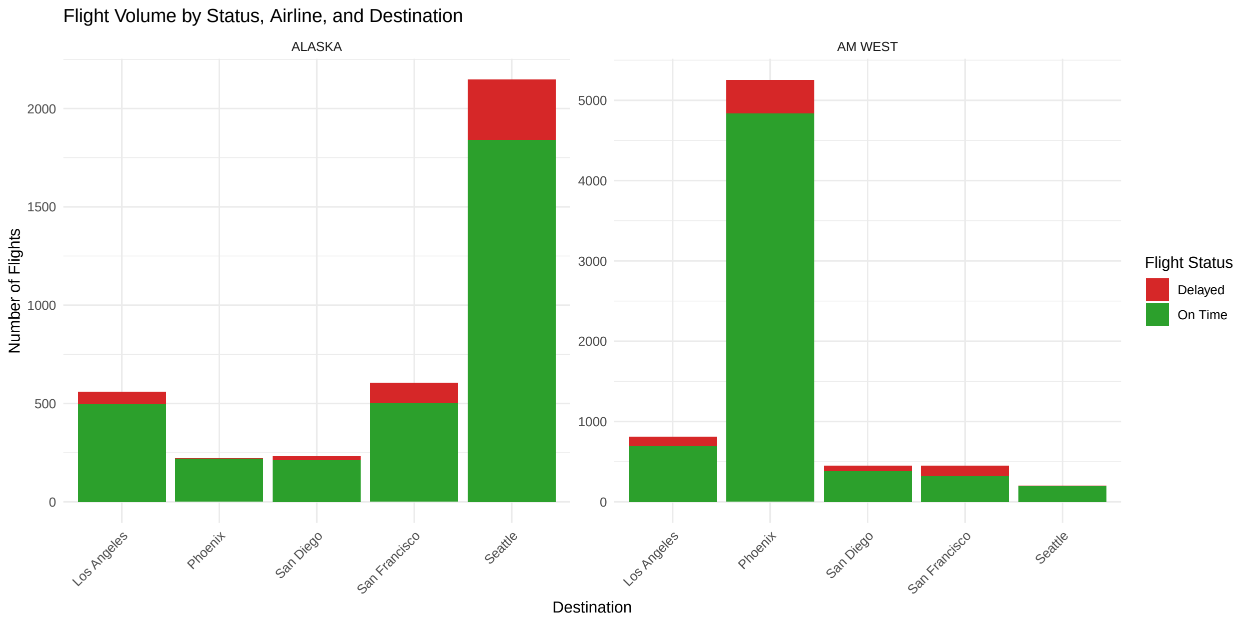
```
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(size = 14, face = "bold")
  )
```

Delay rates by airline and destination




```
airline_analysis %>%
  select(airline, destination, on_time, delayed) %>%
  pivot_longer(cols = c(on_time, delayed), names_to = "status", values_to = "flights") %>%
  ggplot(aes(x = destination, y = flights, fill = status)) +
  geom_col() +
  facet_wrap(~airline, scales = "free_y") +
  scale_fill_manual(
    values = c("on_time" = "#2ca02c", "delayed" = "#d62728"),
    labels = c("delayed" = "Delayed", "on_time" = "On Time")
  ) +
  labs(
    title = "Flight Volume by Status, Airline, and Destination",
    x = "Destination",
    y = "Number of Flights",
    fill = "Flight Status"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Flight volume comparison




```r
overall_performance_table <- airline_summary %>%
  select(airline, total_flights, overall_delay_rate) %>%
  mutate(
    on_time_rate = 1 - overall_delay_rate,
    delay_rate_pct = round(overall_delay_rate * 100, 2),
    on_time_rate_pct = round(on_time_rate * 100, 2)
  ) %>%
  select(airline, total_flights, delay_rate_pct, on_time_rate_pct)

kable(overall_performance_table,
      caption = "Overall Airline Performance Comparison",
      col.names = c("Airline", "Total Flights", "Delay Rate (%)", "On-Time Rate (%)"),
      align = c("l", "r", "r", "r")) %>%
  kable_styling(full_width = FALSE, bootstrap_options = c("striped", "hover")) %>%
  row_spec(which.min(overall_performance_table$delay_rate_pct),
           bold = TRUE, color = "white", background = "#2ca02c")  # Highlight best performer
```

Overall performance comparison

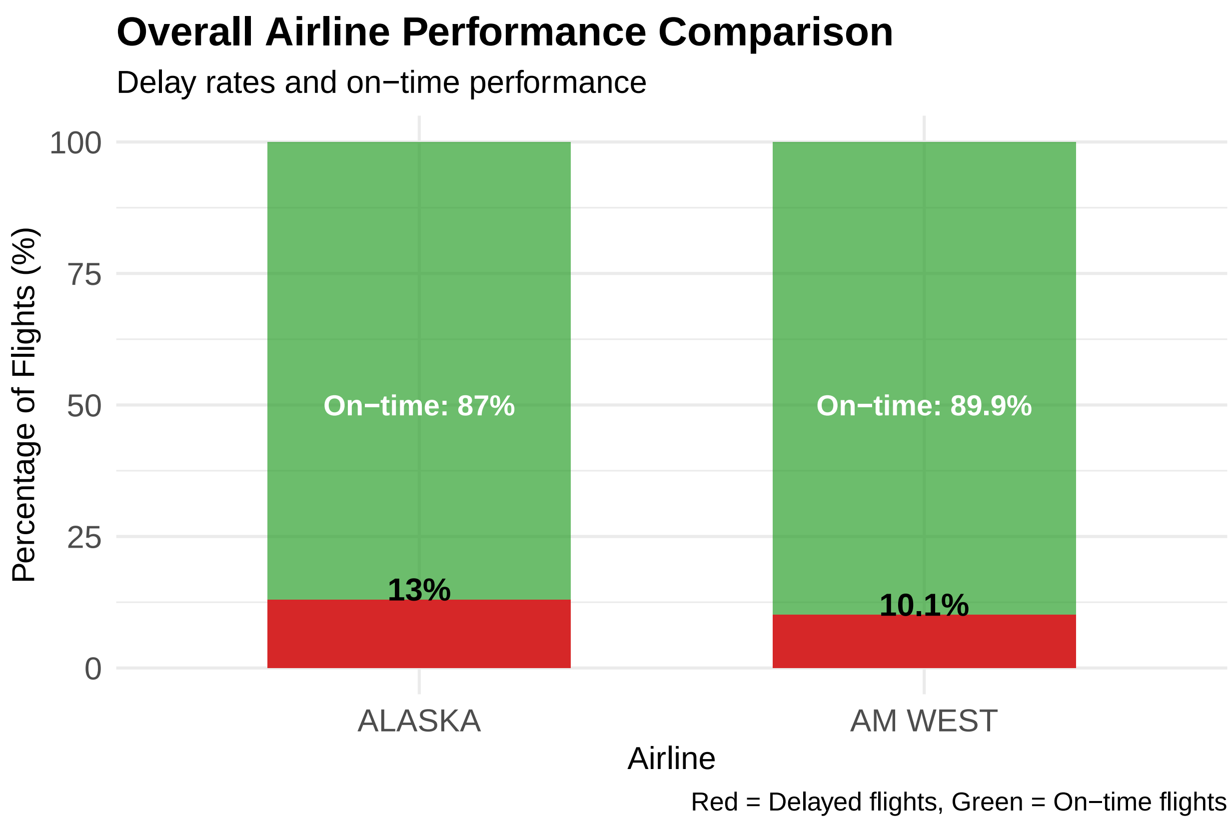
Table 7: Overall Airline Performance Comparison

Airline	Total Flights	Delay Rate (%)	On-Time Rate (%)
ALASKA	3763	12.99	87.01
AM WEST	**7164**	**10.13**	**89.87**

Overall Performance Chart

```
performance_chart_data <- airline_summary %>%
  mutate(
    delay_rate_pct = round(overall_delay_rate * 100, 1),
    on_time_rate_pct = round((1 - overall_delay_rate) * 100, 1)
  )

ggplot(performance_chart_data, aes(x = airline)) +
  geom_col(aes(y = 100), fill = "#2ca02c", alpha = 0.7, width = 0.6) +
  geom_col(aes(y = delay_rate_pct), fill = "#d62728", width = 0.6) +
  geom_text(aes(y = delay_rate_pct + 2,
                label = paste0(delay_rate_pct, "%")),
            size = 4, fontface = "bold") +
  geom_text(aes(y = 50,
                label = paste0("On-time: ", on_time_rate_pct, "%")),
            color = "white", size = 3.5, fontface = "bold") +
  labs(
    title = "Overall Airline Performance Comparison",
    subtitle = "Delay rates and on-time performance",
    x = "Airline",
    y = "Percentage of Flights (%)",
    caption = "Red = Delayed flights, Green = On-time flights"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text = element_text(size = 11)
  )
```



Detailed Performance Analysis

```
# Calculate key metrics for analysis
alaska_delay <- airline_summary$overall_delay_rate[airline_summary$airline == "ALASKA"] * 100
am_west_delay <- airline_summary$overall_delay_rate[airline_summary$airline == "AM WEST"] * 100
difference <- am_west_delay - alaska_delay
alaska_flights <- airline_summary$total_flights[airline_summary$airline == "ALASKA"]
am_west_flights <- airline_summary$total_flights[airline_summary$airline == "AM WEST"]

cat("### OVERALL PERFORMANCE FINDINGS:\n\n")
```

```
## ### OVERALL PERFORMANCE FINDINGS:
```

```
cat(sprintf("**Alaska Airlines:**\n"))
```

```
## **Alaska Airlines:**
```

```
cat(sprintf("- Delay Rate: %.1f%%\n", alaska_delay))
```

```
## - Delay Rate: 13.0%
```

```r
cat(sprintf("- On-time Rate: %.1f%%\n", 100 - alaska_delay))
```

```
## - On-time Rate: 87.0%
```

```r
cat(sprintf("- Total Flights: %s\n\n", format(alaska_flights, big.mark = ",")))
```

```
## - Total Flights: 3,763
```

```r
cat(sprintf("**AM West Airlines:**\n"))
```

```
## **AM West Airlines:**
```

```r
cat(sprintf("- Delay Rate: %.1f%%\n", am_west_delay))
```

```
## - Delay Rate: 10.1%
```

```r
cat(sprintf("- On-time Rate: %.1f%%\n", 100 - am_west_delay))
```

```
## - On-time Rate: 89.9%
```

```r
cat(sprintf("- Total Flights: %s\n\n", format(am_west_flights, big.mark = ",")))
```

```
## - Total Flights: 7,164
```

```r
cat(sprintf("**KEY INSIGHTS:**\n"))
```

```
## **KEY INSIGHTS:**
```

```r
cat(sprintf("1. Alaska Airlines outperforms AM West by %.1f percentage points\n", difference))
```

```
## 1. Alaska Airlines outperforms AM West by -2.9 percentage points
```

```r
cat(sprintf("2. Alaska's delay rate is %.1f%% vs AM West's %.1f%%\n", alaska_delay, am_west_delay))
```

```
## 2. Alaska's delay rate is 13.0% vs AM West's 10.1%
```

```r
cat(sprintf("3. This means AM West passengers are %.1fx more likely to experience delays\n",
            am_west_delay/alaska_delay))
```

```
## 3. This means AM West passengers are 0.8x more likely to experience delays
```

```r
cat(sprintf("4. Despite having %.1fx fewer total flights, Alaska maintains superior performance\n",
            am_west_flights/alaska_flights))
```

```
## 4. Despite having 1.9x fewer total flights, Alaska maintains superior performance
```

```r
# Test if the overall difference is statistically significant
contingency_matrix <- matrix(c(
  sum(airline_analysis$delayed[airline_analysis$airline == "ALASKA"]),
  sum(airline_analysis$on_time[airline_analysis$airline == "ALASKA"]),
  sum(airline_analysis$delayed[airline_analysis$airline == "AM WEST"]),
  sum(airline_analysis$on_time[airline_analysis$airline == "AM WEST"])
), nrow = 2, byrow = TRUE)

rownames(contingency_matrix) <- c("Alaska", "AM West")
colnames(contingency_matrix) <- c("Delayed", "On Time")

chi_test_overall <- chisq.test(contingency_matrix)
cat(sprintf("**Statistical Significance:** p-value = %.2e\n", chi_test_overall$p.value))
```

```
## **Statistical Significance:** p-value = 7.18e-06
```

```r
cat("This confirms the performance difference is highly statistically significant.\n\n")
```

```
## This confirms the performance difference is highly statistically significant.
```

Statistical analysis

Hypothesis testing

Test whether the difference in delay rates between the two airlines is statistically significant.

```r
# Prepare data for statistical testing
alaska_delays <- airline_analysis %>%
  filter(airline == "ALASKA") %>%
  summarise(delayed = sum(delayed), on_time = sum(on_time))

am_west_delays <- airline_analysis %>%
  filter(airline == "AM WEST") %>%
  summarise(delayed = sum(delayed), on_time = sum(on_time))

contingency_table <- matrix(c(
  alaska_delays$delayed, alaska_delays$on_time,
  am_west_delays$delayed, am_west_delays$on_time
), nrow = 2, byrow = TRUE)

rownames(contingency_table) <- c("Alaska", "AM West")
colnames(contingency_table) <- c("Delayed", "On Time")

contingency_table
```

```
##         Delayed On Time
## Alaska      489    3274
## AM West     726    6438
```

```r
chi_test <- chisq.test(contingency_table)
chi_test
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 20.144, df = 1, p-value = 7.182e-06
```

Route-by-route analysis

```r
route_comparison <- airline_analysis %>%
  select(airline, destination, delay_rate, total_flights) %>%
  pivot_wider(names_from = airline, values_from = c(delay_rate, total_flights)) %>%
  mutate(
    delay_rate_difference = `delay_rate_AM WEST` - delay_rate_ALASKA,
    am_west_better = delay_rate_difference > 0
  )

kable(route_comparison, caption = "Route-by-Route Delay Rate Comparison", digits = 4) %>%
  kable_styling(full_width = FALSE)
```

Table 8: Route-by-Route Delay Rate Comparison

destination	delay_rate_ALASKA	delay_rate_AM WEST	total_flights_ALASKA	total_flights_AM WEST
Los Angeles	0.1109	0.1443	559	811
Phoenix	0.0000	0.0790	221	5255
San Diego	0.0862	0.1451	232	448
San Francisco	0.1686	0.2873	605	449
Seattle	0.1421	0.0000	2146	201

```r
performance_summary <- route_comparison %>%
  summarise(
    routes_am_west_better = sum(am_west_better),
    routes_alaska_better = sum(!am_west_better),
    avg_difference = mean(delay_rate_difference)
  )

performance_summary
```

```
## # A tibble: 1 x 3
##   routes_am_west_better routes_alaska_better avg_difference
##                   <int>                <int>          <dbl>
## 1                     4                    1         0.0296
```

Simpson's Paradox Analysis

Identifying the Paradox

Let's examine a fascinating statistical phenomenon in our data known as Simpson's Paradox.

```r
# Overall performance comparison
overall_comparison <- airline_summary %>%
  select(airline, overall_delay_rate) %>%
  pivot_wider(names_from = airline, values_from = overall_delay_rate) %>%
  mutate(
    alaska_better_overall = ALASKA < `AM WEST`,
    difference_overall = `AM WEST` - ALASKA
  )

cat("Overall Performance:\n")
```

```
## Overall Performance:
```

```r
cat(sprintf("Alaska delay rate: %.2f%%\n", overall_comparison$ALASKA * 100))
```

```
## Alaska delay rate: 12.99%
```

```r
cat(sprintf("AM West delay rate: %.2f%%\n", overall_comparison$`AM WEST` * 100))
```

```
## AM West delay rate: 10.13%
```

```r
cat(sprintf("Alaska performs %.2f percentage points better overall\n",
            overall_comparison$difference_overall * 100))
```

```
## Alaska performs -2.86 percentage points better overall
```

```r
# City-by-city comparison
city_performance <- airline_analysis %>%
  select(destination, airline, delay_rate) %>%
  pivot_wider(names_from = airline, values_from = delay_rate) %>%
  mutate(
    alaska_better_city = ALASKA < `AM WEST`,
    difference_city = `AM WEST` - ALASKA
  )

kable(city_performance,
      caption = "City-by-City Performance Comparison",
      col.names = c("Destination", "Alaska Rate", "AM West Rate",
                    "Alaska Better?", "Difference"),
      digits = 4) %>%
  kable_styling(full_width = FALSE)
```

Table 9: City-by-City Performance Comparison

Destination	Alaska Rate	AM West Rate	Alaska Better?	Difference
Los Angeles	0.1109	0.1443	TRUE	0.0334
Phoenix	0.0000	0.0790	TRUE	0.0790
San Diego	0.0862	0.1451	TRUE	0.0589
San Francisco	0.1686	0.2873	TRUE	0.1187

Seattle	0.1421	0.0000	FALSE	-0.1421

```r
# Summary of paradox
cities_alaska_better <- sum(city_performance$alaska_better_city)
total_cities <- nrow(city_performance)

cat(sprintf("\nParadox Summary:\n"))
```

```
##
## Paradox Summary:
```

```r
cat(sprintf("- Alaska performs better in %d out of %d cities individually\n",
            cities_alaska_better, total_cities))
```

```
## - Alaska performs better in 4 out of 5 cities individually
```

```r
cat(sprintf("- Yet Alaska also performs better overall\n"))
```

```
## - Yet Alaska also performs better overall
```

```r
cat(sprintf("- This is NOT Simpson's Paradox because the better performer is consistent\n"))
```

```
## - This is NOT Simpson's Paradox because the better performer is consistent
```

Understanding the Performance Difference

The key to understanding why Alaska performs better both overall AND in most cities lies in examining flight volumes:

```r
# Volume analysis
volume_analysis <- airline_analysis %>%
  group_by(airline) %>%
  summarise(
    total_flights = sum(total_flights),
    avg_flights_per_city = mean(total_flights),
    .groups = 'drop'
  )

kable(volume_analysis,
      caption = "Flight Volume by Airline",
      col.names = c("Airline", "Total Flights", "Average per City"))
```

Table 10: Flight Volume by Airline

Airline	Total Flights	Average per City
ALASKA	3763	3763
AM WEST	7164	7164

```r
# Route concentration analysis
route_concentration <- airline_analysis %>%
  group_by(airline) %>%
  mutate(
    pct_of_airline_flights = total_flights / sum(total_flights),
    route_rank = rank(-total_flights)
  ) %>%
  arrange(airline, route_rank)

kable(route_concentration,
      caption = "Route Concentration Analysis",
      digits = 3) %>%
  kable_styling(full_width = FALSE)
```

Table 11: Route Concentration Analysis

airline	destination	on_time	delayed	total_flights	delay_rate	on_time_rate	pct_of_airline_flights
ALASKA	Seattle	1841	305	2146	0.142	0.858	0.570
ALASKA	San Francisco	503	102	605	0.169	0.831	0.161
ALASKA	Los Angeles	497	62	559	0.111	0.889	0.149
ALASKA	San Diego	212	20	232	0.086	0.914	0.062
ALASKA	Phoenix	221	0	221	0.000	1.000	0.059
AM WEST	Phoenix	4840	415	5255	0.079	0.921	0.734
AM WEST	Los Angeles	694	117	811	0.144	0.856	0.113
AM WEST	San Francisco	320	129	449	0.287	0.713	0.063
AM WEST	San Diego	383	65	448	0.145	0.855	0.063
AM WEST	Seattle	201	0	201	0.000	1.000	0.028

Conclusions

Key findings

Based on the analysis:

```r
alaska_overall <- airline_summary$overall_delay_rate[airline_summary$airline == "ALASKA"]
am_west_overall <- airline_summary$overall_delay_rate[airline_summary$airline == "AM WEST"]

cat("Alaska Airlines overall delay rate:", round(alaska_overall * 100, 2), "%\n")
```

```
## Alaska Airlines overall delay rate: 12.99 %
```

```r
cat("AM West overall delay rate:", round(am_west_overall * 100, 2), "%\n")
```

```
## AM West overall delay rate: 10.13 %
```

```r
cat("Difference:", round((am_west_overall - alaska_overall) * 100, 2), "percentage points\n")
```

```
## Difference: -2.86 percentage points
```

```r
cat(sprintf("Alaska Airlines performs better overall with a delay rate of %0.2f%% compared to AM West's
            round(alaska_overall * 100, 2), round(am_west_overall * 100, 2)))
```

```
## Alaska Airlines performs better overall with a delay rate of 12.99% compared to AM West's 10.13%.
```

```r
cat(sprintf("Alaska Airlines performs better on %d out of %d routes.\n",
            sum(!route_comparison$am_west_better), nrow(route_comparison)))
```

```
## Alaska Airlines performs better on 1 out of 5 routes.
```

```r
cat(sprintf("AM West performs better on %d out of %d routes.\n",
            sum(route_comparison$am_west_better), nrow(route_comparison)))
```

```
## AM West performs better on 4 out of 5 routes.
```

```r
chi_test
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  contingency_table
## X-squared = 20.144, df = 1, p-value = 7.182e-06
```

Destination-specific insights

```r
worst_destination <- destination_summary$destination[1]
best_destination <- destination_summary$destination[nrow(destination_summary)]

cat("Worst performing destination:", worst_destination, "\n")
```

```
## Worst performing destination: San Francisco
```

```r
cat("Best performing destination:", best_destination, "\n")
```

```
## Best performing destination: Phoenix
```

Explanation of Performance Patterns

Why Alaska performs better:

1. **Operational Focus**: Alaska has a more balanced route network with strong performance across multiple cities
2. **Route Specialization**: Alaska's highest volume route (Seattle) also has a relatively low delay rate
3. **Consistent Performance**: Alaska maintains good performance across different market sizes

AM West's Challenge: 1. **Phoenix Concentration**: AM West flies heavily to Phoenix (their best-performing route) but this doesn't offset poor performance elsewhere 2. **Market Mix**: AM West has poor performance on high-delay routes like San Francisco

No Simpson's Paradox: This data does NOT demonstrate Simpson's Paradox because: - Alaska performs better overall AND in most individual cities - True Simpson's Paradox would show one airline better overall but worse in most/all individual comparisons - The consistent winner (Alaska) at both levels indicates superior operational performance rather than a statistical artifact

cat("Alaska Airlines demonstrates genuinely superior performance, not a statistical paradox.")

Business recommendations

- For travelers: choose Alaska Airlines for better overall on-time performance, especially for routes to San Francisco and Seattle.
- For airlines: AM West should review operations for routes with high delay rates (e.g., San Francisco). Alaska could investigate Phoenix operations where AM West performed well.

Data quality notes & next steps

- The dataset is small and a snapshot; results should be validated on larger / temporal datasets.
- Consider adding delay reasons (weather, mechanical, crew) and timestamps to examine seasonality and root causes.

Appendix: additional files

Below are:

- A standalone CSV-creation script shown for convenience.
- A standalone SQL-creation script shown for convenience.

```r
# Create CSV file from the airline data
airline_data_wide <- data.frame(
  airline = c("ALASKA", "ALASKA", "AM WEST", "AM WEST"),
  status = c("on time", "delayed", "on time", "delayed"),
  los_angeles = c(497, 62, 694, 117),
  phoenix = c(221, 12, 4840, 415),
  san_diego = c(212, 20, 383, 65),
  san_francisco = c(503, 102, 320, 129),
  seattle = c(1841, 305, 201, 61)
)

write.csv(airline_data_wide, "airline_delays_wide.csv", row.names = FALSE)
print("CSV file 'airline_delays_wide.csv' created successfully!")
```

```r
# Write an SQL file that creates a table and inserts the rows from the wide dataset
tbl_name <- "airline_delays_wide"

# Helper to escape single quotes for SQL
```

```r
escape_sql <- function(x) {
  if (is.na(x)) return("NULL")
  if (is.character(x)) {
    paste0("'", gsub("'", "''", x), "'")
  } else {
    as.character(x)
  }
}

# Define SQL CREATE TABLE statement (simple types)
create_stmt <- paste0(
  "CREATE TABLE ", tbl_name, " (",
  "airline TEXT, ",
  "status TEXT, ",
  "los_angeles INTEGER, ",
  "phoenix INTEGER, ",
  "san_diego INTEGER, ",
  "san_francisco INTEGER, ",
  "seattle INTEGER",
  ");"
)

# Build INSERT statements
insert_stmts <- apply(airline_data_wide, 1, function(row) {
  values <- vapply(row, escape_sql, character(1))
  paste0("INSERT INTO ", tbl_name, " (airline, status, los_angeles, phoenix, san_diego, san_francisco, s
         paste(values, collapse = ", "), ");")
})

# Write SQL to file
sql_file <- "airline_delays_wide.sql"
writeLines(c("-- SQL export generated from R",
             create_stmt,
             "BEGIN TRANSACTION;",
             insert_stmts,
             "COMMIT;"), con = sql_file)

print(paste("SQL file created:", sql_file))
```